Advanced Data Analysis and Visualization in Logistics

Company: SwiftRoute Logistics Pvt. Ltd.

Project: Logistics Data Analysis Using Python

Task: Week 3 — Advanced Data Analysis and Visualization

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

Dataset: Hypothetical Logistics Shipment Dataset

Dataset Size: 500 simulated shipment records

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1. Introduction

Logistics companies need reliable data to monitor delivery performance, control transportation costs, and improve operation efficiency. Data analysis and visualization help identify patterns, compare performance, and support informed business decisions.

This project focuses on SwiftRoute Logistics Pvt. Ltd., a hypothetical logistics company. A simulated dataset containing 500 shipment records was created and analysed using Python. The dataset includes delivery time, distance, shipment weight, shipment volume, transportation cost, delivery zone, and delivery status.

Exploratory Data Analysis (EDA) was performed using Pandas and NumPy to calculate summary statistics and examine the dataset. Matplotlib and Seaborn were used to create visualizations showing delivery-time distributions, transportation costs, delivery status, zone-level performance, and correlations between numerical variables.

The project aims to demonstrate how data analysis can help identify potential cost drivers, delivery delays, and areas requiring further investigation. Since the dataset is simulated, the findings illustrate analytical techniques rather than actual company performance.

2. Project Objectives

- Create a hypothetical logistics dataset containing 500 shipment records.
- Perform exploratory data analysis using Python.
- Calculate mean, median, standard deviation, distributions, and correlations.
- Create visualizations to examine delivery times, transportation costs, shipment status, and zone-level performance.
- Calculate key logistics performance indicators, including delivery success rate, delay rate, and average transportation cost.
- Identify potential operational bottlenecks and cost patterns.
- Recommend improvements based on the analysis while acknowledging the limitations of simulated data.

3. Dataset Description

A hypothetical logistics shipment dataset containing 500 simulated records was creayed using Python. The dataset represents shipment activities for SwiftRoute Logistics Pvt. Ltd. and contains nine clumns covering shipment identification, delivery performance, shipment characterstics, transportation cost, and delivery status.

Column Name	Description
Shipment_ID	Unique identifier for each shipment
Order_Date	Date the order was placed
Distance_KM	Shipment distance in kilometres
Shipment_Weight_KG	Weight of the shipment in kilograms
Shipment_Volume	Number of shipment units represented
Delivery_Time_Days	Simulated delivery duration in days
Delivery_Zone	North, South, East, or West
Delivery_Status	Delivered, Delayed, or Failed
Transportation_Cost	Simulated cost of transporting the shipment

The dataset was generated for demonstrating purposes. It does not contain actual operational records for SwifRoute Logistics.

4. Methodology

The project followed a structured data analysis process using Python in Google Colab.

Step 1: Data Simulation

A hypothetical dataset containing 500 shipment records was generated using Pandas and NumPy. Random values were used to represent shipment charactertics, delivery times, delivery zones, and delivery status.

Step 2: Data Inspection

The dataset was inspected using head(), info(), shape, and missing-value and duplicate checks. These methods helped examine the structure and basic quality of the data.

Step 3: Exploratory Data Analysis

Descriptive statistics, including mean, median, and standard deviation, were calculated for numerical logistics variables. Correlation analysis was used to examine relationships between distance, delivery time, shipment weight, shipment volume, and transportation cost.

Step 4: Data Visualization

Matplotlib and Seaborn were used to create histograms, bar charts, and a correlation heatmap. These visualizations helped compare shipment outcomes, examine cost distributions, and explore differences between delivery zones.

Step 5: KPI Calculation and Interpretation

Delivery success rate, delay rate, and average transportation cost were calculated. Zone-level performance was also compared to identify patterns that could warrant further investigation.

Step 6: Recommendations

The findings were used to develop recommendations for further logistics analysis. Since the dataset is simulated, recommendations are preliminary and should be validated using actual operational data before implementation.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the structure, distributions, and relationships within the simulated logistics dataset. Python libraries, including Pandas, NumPy, Matplotlib, and Seaborn, were used to examine shipment characteristics and logistics.

5.1 Dataset Inspection

The dataset contains 500 simulated shipment records and nine columns. The dataset was inspected using Python to check its dimensions, column names, datatypes, missing values, and duplicate records. These checks help confirm that the data is structured appropriately for further analysis.

Actual Results from Python

- Total shipment records: 500
- Total columns: 9
- Missing values: 0
- Duplicate records: 0

Interpretation: The dataset is complete with respect to missing values and duplicate records. Therefore, no-missing value treatment or duplicate removal was required for this simulated dataset before performing the planned exploratory analysis.

5.2 Descriptive Statistics

Descriptive statistics were calculated for delivery time, distance, shipment weight, shipment volume, and transportation cost using Python to understand the central tendency and variability of the simulated logistics data. The results are presented below.

Logistics Metric	Mean	Median	Standard Deviation
Delivery Time (Days)	4.48	4.00	2.25
Distance (KM)	259.89	261.00	139.43

Shipment Weight (KG)	50.71	52.08	28.24
Shipment Volume	10.40	10.00	5.69
Transportation Cost	1500.78	1492.96	709.46

Interpretation

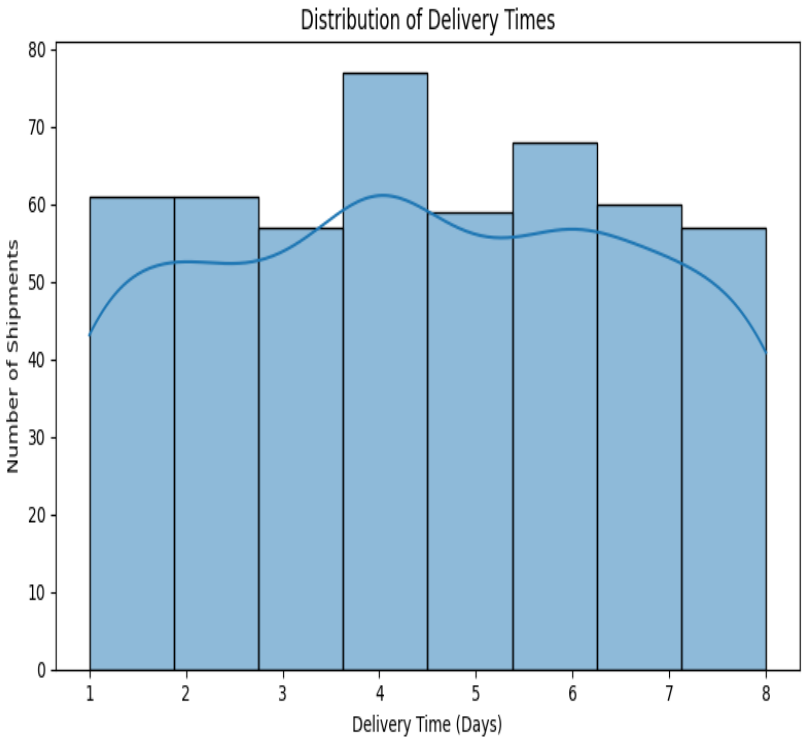
- **Delivery time:** The average delivery time was 4.48 days, while the median was 4 days. This indicates that the typical simulated shipment took approximately four days.
- **Distance:** The average shipment distance was 259.89 km, with a standard deviation of 139.43 km, showing considerable variation in shipment distances.
- **Shipment weight:** The average weight was 50.71 kg, and the median was 52.08 kg.
- **Shipment volume:** The average shipment volume was 10.40, with a median of 10.
- **Transportation cost:** The average transportation cost was 1500.78, and the median was 1492.96. The standard deviation of 709.46 indicates substantial variation in simulated transportation costs.

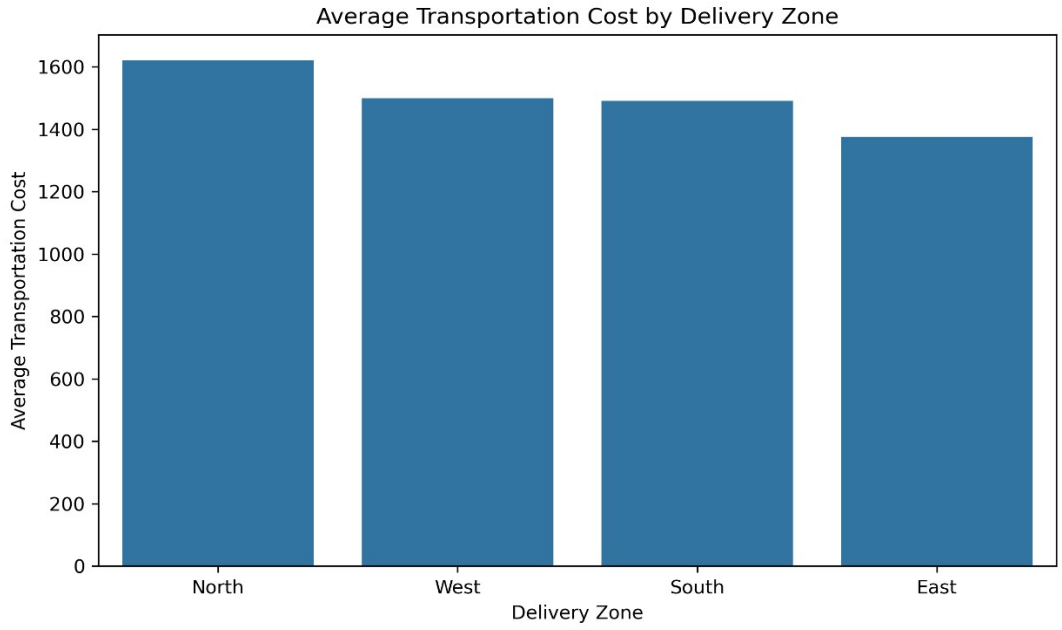
These results summarize the simulated dataset. They should not be interpreted as actual SwiftRoute Logistics performance figures.

5.3 Distribution Analysis

Histograms were created to examine the distributions of delivery times and transportation costs. The delivery-time histogram shows how frequently different delivery durations occur, while the transportation-cost histogram illustrates the variation in simulated shipment costs.

These visualizations help identify common values, variability, and potential unusual observations in the dataset.



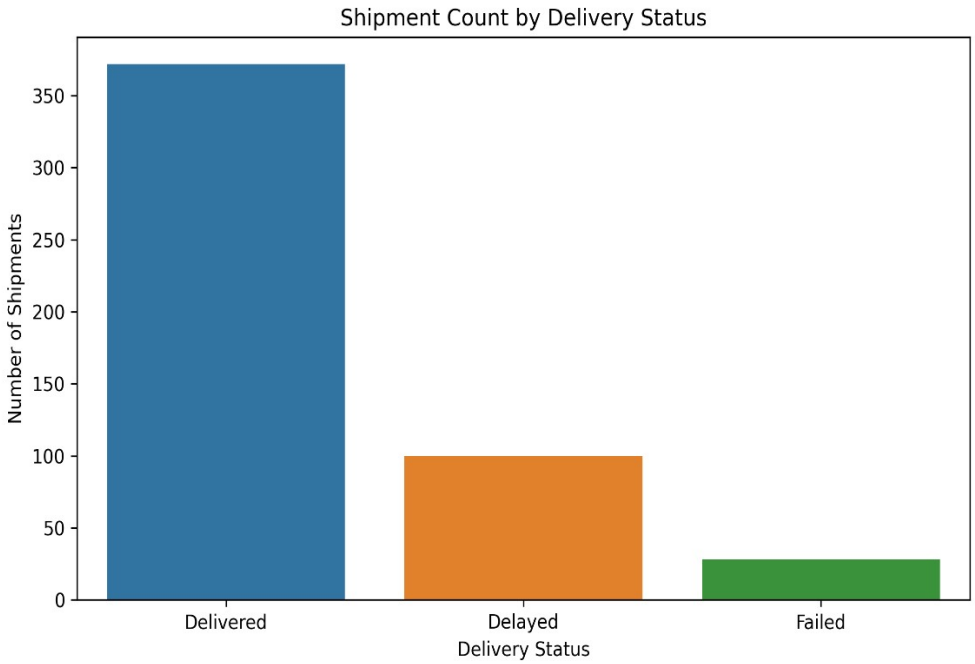


The average transportation cost was compared across the four delivery zones. North had the highest average cost at 1621.92, followed by West at 1499.71, South at 1491.03, and East at 1375.38.

The difference suggests that transportation costs vary across zones in the simulated dataset. North may warrant further investigation to determine whether distance, shipment weight, or other factors contribute to its higher average cost. The results alone do not establish the cause of the difference.

5.4 Delivery Status Analysis:

A bar chart used to compare number of shipments classified as Delivered, Delayed, and Failed. This visualization provides an overview of shipment outcomes and supports the calculation of delivery success and delay rates.



Actual Results and Interpretation:

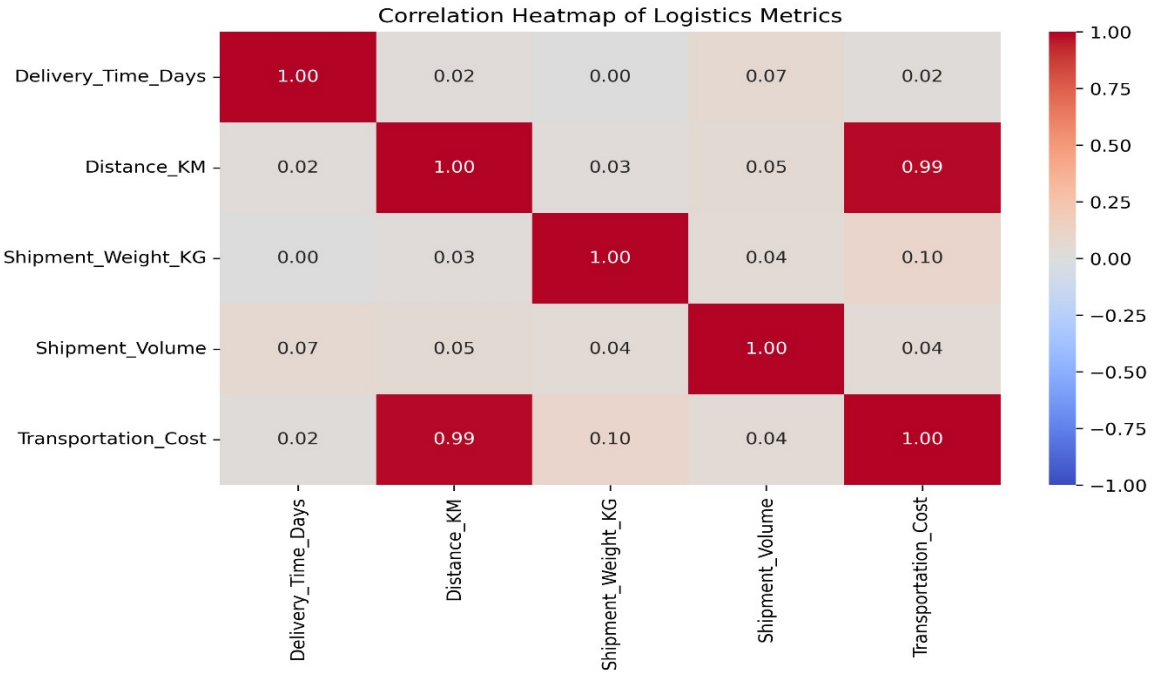
The dataset contains 500 simulated shipments. Of these, 372 were delivered, 100 were delayed, and 28 failed. The delivery success rate was 74.40%, while the delay rate was 20.00%.

These results show that delayed and failed shipments represent areas worth investigating in the simulated scenario. Further analysis could

examine delivery zones, distance, and shipment characteristics to understand possible patterns. Since the dataset is simulated, these figures do not represent actual SwiftRoute Logistics performance.

5.5 Correlation Analysis:

A correlation heatmap was created to examine relationships between delivery time, distance, shipment weight, shipment volume, and transportation cost. The correlation coefficients range from -1 to +1, with values closer to +1 indicating a stronger positive linear relationship.



Actual Results and Interpretation

- **Distance and transportation cost (0.99):** These variables show a very strong positive correlation. In the simulated dataset, longer distances are strongly associated with higher transportation costs. This is consistent with the cost-generation formula, which directly uses distance.
- **Shipment weight and transportation cost (0.10):** This indicates a weak positive linear relationship.
- **Delivery time and distance (0.02):** This indicates almost no linear relationship in the simulated data.

- **Delivery time and transportation cost (0.02):** This also indicates almost no linear relationship.
- **Shipment volume and the other variables (0.04–0.07, except as shown in the heatmap):** These values indicate very weak positive linear relationships.

Overall, the heatmap indicates that distance is the strongest correlate of transportation cost in this simulated dataset. Delivery time was generated independently of distance, which helps explain the near-zero correlation. Correlation does not prove causation, and these results should not be treated as actual SwiftRoute Logistics performance.

5.6 Delivery Performance by Zone:

Delivery performance was compared across the four delivery zones using total shipments, average delivery time, average transportation cost, and delay rate.

Delivery_Zone	Total_Shipments	Average_Delivery_Time	Average_Transportation_Cost	Delayed_Shipments	Delay_Rate_Percent
East	113	4.75	1375.38	24	21.24
North	129	4.19	1621.92	24	18.60
South	136	4.38	1491.03	29	21.32
West	122	4.66	1499.71	23	18.85

Interpretation:

South recorded the highest number of delayed shipments (29) and the highest delay rate (21.32%). East had the longest average delivery time at 4.75 days, while North had the highest average transportation cost at 1621.92. West had the lowest delay rate at 18.85%.

These results suggest that South and East may warrant further investigation into delivery delays and delivery times, while North's higher average transportation cost could be examined alongside shipment distance and weight. Because the dataset is simulated,

these patterns demonstrate analytical methods rather than actual SwiftRoute Logistics performance.